

H-60 Phase 1 — Base-Class Consolidation: Detailed Plan

Prepared: 2026-08-17 · **Updated:** 2026-09-06 (**as-built rewrite:** the asset/config split executed inside Phase 1 — see the end-state note in §2 and the as-built chain in §4; the 2026-08-29 MEDEVAC-to-UH60 ruling is superseded) · **Basis:** master improvement plan §Phase 1 (corrected Snow plan) + §Phase 2.5 (asset/config split, as-built record), the verified UH60↔uh60_config structural diff (2026-08-17), and Riverman's rulings on the Slick armament, MLASS lifetime, and HH60 separation. **Prerequisite:** Phase 0 — **COMPLETE**. Shipped as Stable 0.7.9 (2026-09-04); all Phase 0 PRs merged to Main. **Phase 1 kickoff: GO given 2026-09-05** — gating 0.7.10-cycle config PRs merged (#591/#595/#599/#601; #594/#597 are SQF-only and don't gate), PR strategy ruled (§8.3: sequential PRs).

1. Objective

Today `vtx_H60_base` is **fully declared in four addons** (UH60, uh60_config, H60_SFX, ace_viv) and three variants (`vtx_UH60M`, `vtx_UH60M_SLICK`, `vtx_HH60`) are **declared twice** (UH60 + uh60_config). Whoever loads last silently wins every property collision. This is how the Slick's minigun visibility got inverted without anyone noticing, and it is the leading suspect for issue **#510** (cockpit sound attenuation loss).

Phase 1 ends with **one full declaration per class, everywhere** — the rule both reference projects (AH-64D, MH-47G) enforce without exception:

- Every other touchpoint becomes an honest **delta**: forward-declare the parent, re-open the class, add only your own members.
- Behavior after the change is **provably identical** to before (config-dump diff), except where a divergence is resolved deliberately and on the record.

Phase 1 does **not** merge addons, rename classes, move assets, or delete dead code (that's Phase 2). It is a declaration-ownership cleanup plus one structural move: the HH60's departure into its own addon.

2. Target end-state

Addon	Role after Phase 1 (<i>as-built, 2026-09-06</i>)
uh60_config	Sole owner of all vehicle config (ships later as <code>hct_h60_config</code>): <code>vtx_H60_base</code> + the shared include tree (turrets, parts, AnimationSources, weapons/magazines/sounds, MFD pedestal screens, CfgMoves, fonts declaration, editor subcategory), the crew units, and the sole declarations of <code>vtx_UH60M</code> , <code>vtx_UH60M_SLICK</code> , <code>vtx_UH60M_MEDEVAC</code> , <code>vtx_S70M</code> — one .hpp per vehicle under <code>config/vehicles/</code>
UH60	Asset-only: p3ds + model.cfg, textures/rvmats, sounds, crew anim .rtm files, Font files, stringtable, XEH boilerplate, CBA version check, stub <code>CfgPatches (units[] = {})</code>
HH60 (<i>new addon — addons/HH60, named per Riverman 2026-08-29; CfgPatches vtx_hh60</i>)	The Air Force bird : sole owner of <code>vtx_HH60</code> (config-only; flies the shared p3d until the HH-60W model conversion)
H60_SFX	Sound delta on <code>vtx_H60_base</code> (audit found its block already sound-only in content — no demotion edits were needed; gained the <code>vtx_hh60</code> edge for its Compat patch)
ace_viv	Unchanged except the <code>vtx_UH60_config</code> edge (still a one-property delta; see §6 warnings)

Addon	Role after Phase 1 (<i>as-built, 2026-09-06</i>)
MH60M / MH60S	Keep their own vehicle declarations (ruled 2026-09-05) as deltas deriving from the owner's base; <code>requiredAddons</code> corrected to <code>vtx_UH60_config</code> ; their includes re-pointed to <code>uh60_config</code> 's tree
Other re-openers (mfd, fms, anvishud, sfmplus, acre, cas, hoist, misc, weapons, ui, aar, doorguns)	Unchanged in content — each declares an honest <code>vtx_UH60_config</code> edge where it re-opens owned classes

End-state note (as-built, 2026-09-06): the AH-64D-style **asset/config split** ruled on 2026-09-05 (Riverman + BroBeans) was executed *inside* Phase 1, not staged for later. The original plan gated it behind an MFD-wiring inversion (`uh60_config` was the last loader and supposedly could not own declarations deriving from later-loading screen templates); BroBeans' load-order experiment (2026-09-05, WP0 harness) disproved the gate — **rapified external inheritance resolves by name at config read time**, so the owner loads early and the screen addons follow. WP1 staged single ownership in UH60 (#603) exactly as planned below; PRs A/B/C (#607/#608/#609) then moved everything to `uh60_config`. The WP texts below are preserved as written for the record; where a WP says "UH60 becomes the single source," the final owner is `uh60_config`. See the master plan §Phase 2.5 for the full as-built record.

Variant identity rulings that govern the merge:

- **`vtx_UH60M_SLICK` is the armed ViV-cargo variant.** `uh60_config`'s values (miniguns **shown**) are the intent; UH60's hide-lines are the wrong copy and are deleted. (*Riverman, 2026-08-17*)
- **`vtx_UH60M_MEDEVAC` (HH-60M) is the only unarmed military variant.** `vtx_S70M` is civilian, unarmed.
- **`vtx_HH60` is currently a G-dressed airframe with a W-style cockpit; its long-term identity is the HH-60W.** No G-specific fidelity work; the class name does not change in Phase 1 (rename decision deferred to the model conversion — consider a scope-1 compat alias then).
- **MLASS stays flyable** until its planned removal; Phase 1 must not degrade it.

3. Work packages, in order

WP0 — Verification harness (build first, use throughout)

1. **Config-dump script:** for each spawnable class, dump every property with **values** (the master plan's snippet — `configProperties` with `getNumber` / `getText` / `getArray`; the Snow plan's names-only version misses silent value changes).
2. **Baseline capture on Main** for the full spawn matrix: `vtx_UH60M`, `vtx_UH60M_SLICK`, `vtx_UH60M_MEDEVAC`, `vtx_HH60`, `vtx_S70M` (*the Snow plan omitted it*), `vtx_MH60M`, `vtx_MH60M_DAP`, `vtx_MH60M_DAP_MLASS`, and the MH60S set.
3. Store dumps as files; the acceptance test for every WP below is `diff old new` = empty except the deliberate changes listed in that WP.

WP1 — `vtx_H60_base`: one owner

1. UH60's declaration ([UH60/config/cfgVehicles.hpp:38-458]) becomes the single source.
2. **H60_SFX** (config.cpp:200-312): demote to a delta carrying *only* its sound content (`soundEngineOn*`, `class sounds`, `SoundSets`). Two traps from the corrected plan: - `playerPosition = 1` (config.cpp:197) sits on the **vanilla** `Heli_Transport_01_base_F` → `CargoTurret` — it must **not** move into `vtx_H60_base`. - H60_SFX redeclares **76** `vtx_*` classes in total; sweep `Compat.hpp` siblings into delta form in the same pass.
3. **ace_viv**: stays exactly as-is (see §6 — do not "clean up" its GVAR macro).

4. **Base-level value collisions** — resolve to the later-loading (live) value, then delete the loser: - RotorHFOld: keep `animPeriod=5; initPhase=0` (uh60_config), delete UH60's `1 / -0.125`. - The ~60 duplicated AnimationSources between the two AnimationSources blocks: keep one copy of each; uh60_config retains only sources it genuinely adds or extends (`mass=`, `forceAnimate`, `onPhaseChanged`, `blade-fold set`, `windows`, `wipers`).
5. uh60_config's `vtx_UH60_base` delta keeps its legitimate content: `class Armory` blank, `TextureSources` (12 liveries), `hct_turret_1/2/3`, its class EHs and CBA functions. *(The corrected plan explicitly rejects making it a bare stub — delta-extension from a later loader is the reference-blessed pattern.)*

Deliberate changes to log: none expected at runtime — WP1 should be config-dump **byte-identical** for every variant.

WP2 — Army variant merges (`vtx_UH60M`, `vtx_UH60M_SLICK`)

1. **vtx_UH60M**: the two halves are disjoint (identity/visual in UH60; seats/MFD/medical in uh60_config). Keep the full declaration in UH60; uh60_config's block remains as the delta that wires MFD elements (it must stay in uh60_config — it references `uh60_mfd/fms/anvishud` content that loads after UH60). Net config: identical.
2. **vtx_UH60M_SLICK**: merge per the armament ruling — miniguns/mounts/gunner seats **shown**. Delete UH60's five `ANIM_INIT(...,1)` hide lines (UH60:564-568); keep uh60_config's `initPhase=0` set and granular `CabinSeats_1/2/3`. Remove the now-dead master `CabinSeats_Hide,1` and `Hoist_hide/sight` lines only if the config dump proves them inert; otherwise keep and log.
3. **Audit the Slick's turret-lock indexes**: uh60_config's `cabindoor_L/R onPhaseChanged` locks turrets `[[3],[4]]/[[1],[2]]`, which matches neither HH60 (`[[5],[6]]`) nor MEDEVAC mappings. Verify against the merged Slick's actual `Turrets` order (UH60's block contributes `cargoTurretsDoor.hpp` — 7 door cargo turrets) and correct. **This is a deliberate change if wrong today; test door-lock behavior per seat in-game.**
4. **MEDEVAC placement — RULED (Riverman, 2026-08-29): ~~vtx_UH60M_MEDEVAC~~ moves to UH60 — SUPERSEDED (2026-09-05)** by the asset/config split executing inside Phase 1: with uh60_config the sole owner, the MEDEVAC was already home. As-built it lives in `uh60_config/config/vehicles/UH60M_MEDEVAC.hpp` beside its Army siblings — the 2026-08-29 ruling's intent (family together, one place) is satisfied in the other direction.

Deliberate changes to log: Slick turret-lock indexes (if corrected). Everything else byte-identical — including the relocated MEDEVAC, whose config dump must not change.

WP3 — HH60 separation (new addon)

1. Create `addons/HH60` (folder name ruled by Riverman, 2026-08-29) with a `CfgPatches` class distinct from the vehicle class (`vtx_HH60_addon` or similar — avoid colliding with `vtx_HH60`), `requiredAddons[] = {"vtx_UH60", "vtx_UH60_mfd", "vtx_UH60_fms", "vtx_UH60_anvishud"}` — the same set uh60_config needs, because the HH60 is the only variant wiring the full-FLIR `VTX_MFD_1..4`.
2. Move **both halves** into it: UH60's full declaration (UH60:498-550) + uh60_config's delta (uh60_config:340-401), merged into one clean declaration deriving from `vtx_UH60_base` — the MH60M/MH60S pattern.
3. Resolve its one conflict on the record: `CabinSeats_Hide` → `initPhase=0` (uh60_config's granular replacement is the live behavior; UH60's `,1` line was dead-but-misleading).
4. `units[]`: remove `vtx_HH60` from UH60's `CfgPatches`, add to the new addon's.
5. **Assets stay in UH60** (shared `vtx_uh60.p3d`, `pavehawk.hpp` model.cfg fragment, HH60/HH60_Fire textures) — path references need no dependency edge. The new addon is config-only until the HH-60W model work.
6. **Load-order fix for patchers of vtx_HH60**: `H60_SFX` (`Compat.hpp` adds `CargoTurret_01` to `vtx_HH60`) currently requires only `vtx_uh60 + vtx_uh60_config`. Either add the new HH60 addon to `H60_SFX`'s `requiredAddons`, or move that one compat block into the new addon. Recommendation: add the dependency — keeps sound content in the sound addon.
7. `uh60_config` keeps **nothing** HH60-specific afterward.

Deliberate changes to log: none at runtime — the merged HH60 config must dump identically to today's post-load-order result.

WP4 — requiredAddons and inherit-header cleanup

1. Add "vtx_UH60" to MH60M/config.cpp:9 and MH60S/config.cpp:9 (they derive from vtx_H60_base but never declare the dependency — works today only by accident of load order).
2. CfgAnimationSourcesInherit.hpp (UH60; consumed only by MH60M/MH60S): with single ownership established this forward-declaration crutch may be simplifiable — **investigate, but defer any change to Phase 2/4 if it isn't a pure win**. It is not required for Phase 1's goal.
3. While in the area, note (do not fix) Phase 2 candidates uncovered by the diff: the disabled parallel livery system (vehicleCustomization.hpp, include commented at UH60:87), units[] case mismatch vtx_UH60M_Slick vs SLICK (harmless), dead doorguns.hpp.

4. Branch & PR strategy

The master plan scoped Phase 1 as one PR; with the HH60 amendment, three **sequential** PRs keep each diff reviewable and independently regression-tested, at the cost of the middle state shipping nothing user-visible:

1. restructure/base-single-owner — WP1 (base class + H60_SFX demotion). Config-dump proof: byte-identical.
2. restructure/army-variant-merge — WP2. Proof: byte-identical except logged Slick items.
3. restructure/hh60-addon — WP3 + WP4. Proof: byte-identical per variant; spawn matrix green; #510 retest.

As-built chain (2026-09-05/06) — WP1 shipped as planned; the rest restructured when the asset/config split moved inside Phase 1 (Riverman's ruling after the load-order experiment):

1. **#603** restructure/base-single-owner — WP1 as planned. Proof: byte-identical (580,227 dump lines both sides).
2. **PR A, #607** restructure/config-owner-variants — WP2 executed as merges *into uh60_config* (one .hpp per vehicle under config/vehicles/) + the dependency-edge flip. Proof: 0 value changes, 65 inert classOrder shifts. Slick armament ruling applied (miniguns shown, five dead hide-lines dropped); WP2.3's turret-lock audit deferred to the in-game soak.
3. **PR B, #608** restructure/hh60-addon — WP3 as planned (CfgPatches vtx_hh60, renamed from vtx_HH60_addon on BroBeans' review — see §8.1; CabinSeats_Hide → live initPhase=0 resolved on the record; H60_SFX gains the edge).
4. **PR C, #609** restructure/uh60-asset-only — the base + full config-tree move to uh60_config; UH60 reduced to the asset stub. (The original plan deferred this to §Phase 2.5.)
5. **PR D** restructure/phase1-tidy — dead-file deletion, dependency audit (uh60_doorguns edge), CfgAnimationSourcesInherit documented as rapify scaffold, this docs catch-up.

Same workflow as Phase 0: branch published at creation, commits local until Riverman verifies, PR on his word. Each PR's build = affected PBOs minimum (scons all = 26 PBOs with the HH60 addon) plus the config-dump diff. Test cadence ruled mid-phase (Riverman, 2026-09-05): merge the chain, push one dev build with everything, fix issues as they surface — the per-PR soak plan below is superseded.

Coordination with the 0.7.10 dev cycle (added 2026-09-04): several feature/fix PRs are in flight against the same config territory Phase 1 restructures — #591 (FMS interaction points), #594 (FLIR slew deadzone), #595 (ACRE comms sync), #597 (network improvements), #599 (MH60M door initPhase). Sequence at kickoff: either land/close them first, or capture the WP0 baseline *after* each merge and rebase the restructure branches — a Phase 1 branch must never carry a config-dump diff caused by someone else's merge. Re-capture the baseline whenever Main moves.

5. Verification & acceptance

- **Config-dump diff** (WP0 harness) — the primary proof, run per-variant per-PR.

- **Spawn matrix** — every class in WPO's list spawns in Eden/Zeus, textures and seats correct, no RPT errors. Include getting in pilot/copilot/door-gun/cargo seats.
- **Issue #510 as acceptance test** (*from the master plan; #510 and #439 now sit on the Phase 1 milestone*): cockpit sound attenuation depends on turret-config merges across the four competing base declarations; if Phase 1 is done right, attenuation works (or its true cause is exposed). Test UH60M + MH60M cockpit audio before/after. Second candidate mechanism from the MH-47G reference (2026-08-23): a missing View Geometry LOD silently disables attenuation entirely — check the p3d before concluding the config diff explains everything.
- **Issue #439 retest** (hover symbology on UH/HH models) after WP3 — `vtx_HH60` was one of the duplicate-delta'd classes.
- **Slick functional test**: door guns present and usable (armament ruling), ViV cargo loading works, door-lock behavior per seat (WP2.3).
- **MLASS regression**: spawns, 4 pylons work, FLIR page carets/icons intact (Phase 0 #566 behavior preserved).
- **MP smoke test** on a dedicated server for one Army variant + HH60: crew seats, doors, sounds — the load-order-sensitive surfaces.

6. Do-NOT list (traps recorded in the corrected plan and diff)

1. **Do not move** `playerPosition = 1` (H60_SFX config.cpp:197) into `vtx_H60_base` — it belongs to the vanilla CargoTurret path.
2. **Do not copy** `GVAR(lockSeats)` **text between addons** — in `ace_viv` it expands to `vtx_ace_viv_lockSeats`; pasted elsewhere it expands differently and silently breaks `ace_viv`'s scripts. Leave `ace_viv` untouched.
3. **Do not re-open a populated nested container from a second addon** (the MH-47G "shadow-not-merge law"): re-opening `class Turrets` replaces the whole tree. Deltas add top-level members or extend via forward-declared inner classes only.
4. **Do not resolve collisions toward UH60's values** — the later loader's value is today's live behavior; keeping UH60's would change the aircraft (the Slick incident is the proof).
5. **Do not relocate the MASS_* macros casually** — they live in `uh60_config/script_component.hpp:20-42` and feed its `AnimationSources`; if any animation-source file moves addons, the defines must move to a shared header first. `hct_minigun.hpp` additionally depends on `POS_MOVE_CABIN_COCKPIT` (`main/script_macros.hpp`) — verify include chains before moving it.
6. **Do not touch MLASS-specific content beyond keeping it working** — it is scheduled for removal; don't invest, don't break.
7. **Do not rename** `vtx_HH60` in this phase — mission/save compatibility; the rename rides with the HH-60W model conversion.

7. Risks & mitigations

Risk	Mitigation
Silent value change survives the merge	Value-capturing config dump (not names-only); per-variant diffs attached to PRs
Load-order patcher breaks when <code>vtx_HH60</code> moves (H60_SFX Compat, missions referencing the class)	Class name unchanged; explicit requiredAddons edge; spawn matrix + #439/#510 retests
Slick turret-lock indexes were wrong all along	Treated as deliberate, tested change (WP2.3), not silently "fixed"

Risk	Mitigation
H60_SFX demotion drops a sound property	Its 76-class redeclaration sweep is diffed class-by-class; #510 test doubles as the audio acceptance gate
Regression discovered post-merge	Three small sequential PRs — each independently revertible; bisection stays cheap

8. Open items

1. ~~Name for the new addon folder~~ — **RULED (2026-08-29):** `addons/HH60`; CfgPatches class first shipped as `vtx_HH60_addon`, then **renamed** `vtx_hh60` **on BroBeans' #608 review (2026-09-06)** for house-style parity (`vtx_uh60` / `vtx_mh60m` / `vtx_mh60s`). The plan's "keep it distinct from the vehicle class" caution was unfounded — CfgPatches and CfgVehicles are separate roots, and `vtx_mh60m` / `vtx_MH60M` have coexisted for years.
2. ~~MEDEVAC declaration placement~~ — **RULED (2026-08-29): moves to UH60 SUPERSEDED (2026-09-05):** stayed in `uh60_config`, which became the sole owner (see WP2.4 note and §4 as-built chain).
3. ~~Three-PR split vs single PR~~ — **RULED (Riverman, 2026-09-05): sequential PRs, per §4** (as-built: five — #603 + PRs A–D). Standing policy from here on: phases are planned as sequential PRs unless Riverman rules otherwise for a specific phase.
4. **Still open:** the Slick per-seat door-lock audit (WP2.3) rides the post-merge dev-build soak, alongside the #510/#439 acceptance retests — indexes were preserved as-is through the merges.